REFERENCE · R28

DRIVING CONDITIONS GUIDE

Sand, mud, rock, water, corrugations, altitude, night & steep terrain

2009 LC76 SW · 1HZ 4.2L Diesel · low-boost turbo (0.6 bar) · intercooler
Last updated: 12 October 2026

Basics for any terrain. Thumbs outside the steering-wheel rim — when a front wheel hits a rock or drops into a rut the wheel can kick hard enough to break them (St Pierre White 2014, p. 192; Cobley 2013, ch. 12; Allen 2021, 'Controls and Communications'). Hold the wheel lightly in ruts and sand tracks, and always know which way the front wheels are pointing (Cobley 2013, ch. 11). Seatbelts on for slopes and rough ground; off only for deep wading (§4). In thick bush, windows up to eye height against branches (St Pierre White 2014, p. 187). Walk anything you cannot see the far side of before you drive it (Sheppard 2008, §5.3-18).

1 — SAND

Sand is the most forgiving of the serious terrain types — but only if you read it correctly and arrive with your tyres already deflated. Overconfidence and under-preparation kill more sand runs than anything else. This section covers the 76 specifically: 3,000 kg GVM (with a full-trim loaded mass now measured 280 kg over it — R11 / R29), long wheelbase, and 265/75R16 AT tyres at ~683 kg payload budget.

READING SAND BEFORE YOU ENTER

1. **Colour.**
   Wet or damp sand is usually darker and more compact — better traction. Pale, dry, bleached sand is soft and powdery — worst traction. On beaches, the wet tidal band gives a firm surface; the dry zone above the high-water mark is the danger zone.

2. **Texture and crust.**
   A wind-blown crust feels firm underfoot but will not support the 76's weight. Punch through it with your boot heel before committing. Salt pans with sand overlay behave similarly — soft underneath a deceptive hard cap. See Salt Pans, below.

3. **Moisture cue.**
   Look for vegetation line: dense growth marks higher moisture content and more compact sand. Bare bowls below the vegetation line are typically the softest spots. Track other vehicles — if wheel tracks sink more than 30 mm, drop pressure before entry.

4. **Dune face angles.**
   Windward face = gradual rise, compacted. Leeward face (slip face) = steep, loose, soft. Never attack a slip face at speed. On dune crests, approach slowly — the leeward drop may be steeper than the climb.

5. **Wind direction.**
   Prevailing wind blows sand off the windward face and deposits it on the leeward side. If the wind is in your face as you climb, the slope is firm. If the wind is at your back on a face, you are on the soft slip side.

TYRE PRESSURE FOR SAND — 265/75R16 AT

Tyre pressures are owned by R14 (Tyre Management): its table gives the front and rear pressure for each surface and the speed ceiling that goes with it. Confirm with the Tyre Pressure Calculator before the first sand section — vehicle weight changes the targets. The table below says which R14 row to use for each kind of sand; R14's dune row is the floor (owner ruling, 12 Oct 2026).

SAND TYPE	R14 ROW TO USE	NOTES
Firm beach / wet tidal	Sand (soft, deep)	Below the high-water mark and soft underfoot it behaves like soft sand. Watch sidewall bulge. Beach work at low tide only.
Dry inland track	Sand (soft, deep)	Standard desert track sand. The wider footprint improves flotation markedly on the 76's weight.
Soft inland / dunes	Sand (soft, deep); Sand (extreme — dunes) only if still sinking	Deep, soft or loose sand. Never go below R14's dune row — below it the bead can separate and the sidewall fold under the 76's weight.
Dune slip face	Sand (extreme — dunes)	Last resort, short distances, under 20 km/h. Combine with full momentum commitment. Do not attempt in isolation. Re-inflate in stages afterwards (R14).

MOMENTUM MANAGEMENT

1. **Commit before you enter.**
   Hesitation in sand is the most common cause of bogging. If conditions are marginal, stop before the section, walk the line, deflate to target pressure, then enter with purpose. The time to decide is not while you are moving.

2. **Maintain a steady throttle.**

Not fast — consistent. The turbo 1HZ likes a mid-rev working range in soft sand (not lugging, not over-revving). Smooth, progressive throttle preserves the footprint and avoids wheelspin that digs the vehicle in.

3 **The stall point.**
When forward motion stops or slows dramatically under sustained throttle, you have found the stall point. Immediately ease off — continued wheelspin digs you in at an accelerating rate. The vehicle can recover from a gentle stop; it cannot recover from a buried axle.

4 **Long wheelbase penalty.**
The LC76 SW has a 2,730 mm wheelbase (R25). In cross-dune or uneven sand, the belly clearance advantage of a short-wheelbase vehicle is absent. Stay aware of high-centering risk in soft bowl crossings. See Section 3 (Rock) for break-over angle context.

LINE SELECTION

▸ **Follow existing tracks where possible.**
Compacted by previous vehicles — superior traction — and no new scar on the landscape (St Pierre White 2014, p. 184; no-trace doctrine, R17 §5). Sheppard prefers the firmer unbroken sand beside a churned track (2008, §5.3-29); the library keeps to existing tracks. Where the ruts are deep enough for the middle to touch the diffs, straddle them — wheels on the ridges, or one wheel in and one out — rather than drop both sides in (owner ruling, 12 Oct 2026; St Pierre White 2014, p. 191). Go slowly: the risk when straddling is slipping into the rut (Cobley 2013, ch. 11).

▸ **Ride the crests, not the troughs.**
Crest sand is wind-compacted and drier. Trough sand collects loose material and moisture — softer and more variable. Where possible, keep wheels on high ground between dunes.

▸ **Avoid soft bowls.**
Circular depressions in dune fields collect the finest, loosest sand — almost always the deepest soft spots. Go around them, not through them.

▸ **Approach dune faces straight.**
Any angle less than perpendicular to the face creates a side-loading risk on the 76 — the suspension lifts uphill and the downhill wheels sink. Straight up, straight down.

CORRECTIVE TECHNIQUES

1 **Steering into a sink.**
If the nose starts to dive into soft sand, a small steering correction toward the firmer side (usually upwind) can redirect weight. Do not over-steer — large corrections in soft sand steer the vehicle off-line while the tyres continue forward, causing it to dig sideways.

2 **Power out vs dig in — the decision.**
If momentum is still present, maintain throttle and steer for the firmest available line. If motion has stopped completely, ease off immediately. Applying power from a dead stop in deep sand buries you. The recovery sequence is: ease off → assess → reverse on your existing track if possible → reassess entry.

3 **Reverse out — don't rock.**
Rocking back and forth in sand usually digs the vehicle in deeper (St Pierre White 2014, p. 183). Clear the sand in front of the wheels, reverse along your own tracks — the axles twist the other way and find grip — and try again with a little more speed. If that fails, deploy the recovery gear rather than keep trying (St Pierre White 2014, p. 184; Sheppard 2008, §5.3-28; owner ruling, 12 Oct 2026). Rocking stays a rock and mud technique.

STOPPING, STARTING AND DUNES

▸ **Sand is firmer when it is cool.**
Early morning, evening and after rain the grains pack and carry more weight; the heat of the day after a long dry spell is the worst time (St Pierre White 2014, p. 183).

▸ **Stop on a firm patch, without the brakes.**
Ease off and let the vehicle roll to a stop, ideally pointing slightly downhill. Braking pushes up a wall of sand in front of the tyres and can break the crust (St Pierre White 2014, p. 184; Sheppard 2008, §5.3-29).

▸ **Pull away from your own tracks.**
Reverse about a metre along your own tracks first, then drive off — the vehicle has momentum before it meets the sand wall (St Pierre White 2014, p. 184).

▸ **No gear changes on a dune face.**
Pick the gear on the firm ground before the climb; a change on the slope usually kills the momentum (St Pierre White 2014, p. 184; Sheppard 2008, §5.3-28).

▸ **Stop at the crest.**
Ease off near the top so the vehicle stops level, nose just over the edge, and look at the far side before going over. If you stop still pointing up, reverse straight down and try again (St Pierre White 2014, p. 185).

▸ **Down the slip face straight, in low range.**
Low second keeps the wheels rolling and the nose from digging in (Sheppard 2008, §5.3-26); St Pierre White uses low first on a moderate face and low second on a very steep one (2014, p. 186). If the back starts to come round, accelerate firmly — never brake. On a dune a slide becomes a rollover in an instant (St Pierre White 2014, p. 186).

- **Never cross a dune face sideways.**
  The lower wheels dig in and the vehicle rolls. If a track forces a short traverse and the back breaks away, steer downhill and keep the power on (St Pierre White 2014, p. 186).

- **Deep sand tracks: light hands.**
  The vehicle runs as if on rails. Fighting the wheel crabs the front tyres, wastes power and can spin the vehicle out; guide it gently and be ready to catch a sudden swing (St Pierre White 2014, pp. 186–187).

- **Climbing out of a high-walled track.**
  Ease off, turn the wheel hard and immediately bring it back to nearly straight. Holding full lock is how vehicles roll when the front wheels suddenly bite. If it will not climb out, stop and try in reverse (St Pierre White 2014, p. 187).

SALT PANS

A dry, white pan can look like a road and be a crust over deep, sticky mud. Vehicles are lost on the Botswana pans almost every year, and a bogging can take days to recover from (St Pierre White 2014, pp. 192–193; Sheppard 2008, §5.3-32).

- **Test on foot first.**
  Skirt the edge to find the shortest crossing, then walk ahead of the vehicle. If your boots break the crust, do not drive it. If they don't, dig a hole about 25 cm deep: hard and dry underneath may be safe — but softer patches can lie further on (St Pierre White 2014, p. 192).

- **Stay exactly on the established track.**
  Tracks strengthen the pan beneath them; a metre off can be bottomless (Sheppard 2008, §5.3-32; owner ruling, 12 Oct 2026). St Pierre White suggests driving parallel to fresh tracks (2014, p. 193); the library does not adopt it — it is less safe and leaves new scars, which he condemns himself (p. 272).

- **4WD, diff locks, low range, steady and slow.**
  Low-range third or fourth and no sudden steering — turned wheels plough. Lower tyre pressures help little on a pan. Watch the depth of your own tracks as you go (St Pierre White 2014, pp. 192–193).

- **If it breaks through, decide at once.**
  Either keep the power on to reach firm ground, or stop immediately and reverse out in your own tracks with the wheels straight. Stopping early keeps you close to firm ground and anchor points (St Pierre White 2014, p. 193).

- **Not with one vehicle if it can be avoided.**
  A pan recovery needs a second vehicle with a winch and has to start at once. Don't stop on the pan to look at the view, and no sharp turns at speed — the outside wheels dig in and the vehicle can roll (St Pierre White 2014, p. 193).

RECOVERY PRE-SET BEFORE A SAND SECTION

Before entering any long or serious sand section, the following must already be done — not done when you are stuck:

1. **Tyre pressure deflated to target.**
   Not "I'll drop them if I need to." Drop them before entry. Reflating on the far side is not an emergency.

2. **Recovery boards in reach.**
   Not strapped to a roof rack under a load. Accessible within 30 seconds of stopping.

3. **Front tow point clear.**
   Snatch block and tow strap accessible. Not buried under camp gear.

4. **4WD low range engaged.**
   Engaged before entering soft sand — not attempted mid-section. The part-time transfer case must be engaged while stationary or at very low speed.

5. **Windows up, not cracked.**
   Fine blowing sand enters cabin immediately. Cracking reduces not eliminates ingress. Close fully in active sand sections.

6. **Air filter status confirmed.**
   Check pre-cleaner condition before extended sand running. Cross-ref R4, R18.

CROSS-REFERENCE R9 Vehicle Recovery · R14 Tyre Management · R19 AT Tyre Review · Tyre Pressure Calculator

2 — MUD

Mud is sand's more unforgiving cousin. The variables are harder to read, the consequences of mistakes are more immediate, and the cleaning aftermath cannot be skipped. The 1HZ and the 76's intake path introduce specific risks in thick or splashing mud that do not apply to other terrain types.

MUD TYPES — DIFFERENT PHYSICS, DIFFERENT RISKS

Clay Mud — HIGH RISK

Dense, sticky, rapidly packs tyre treads to a slick surface. Traction disappears fast. Builds up in wheel arches and clogs brakes. Never power through — momentum and line selection are everything. Exit speed matters; do not brake in clay mud.

Black Cotton — EXTREME

Vertisol clay soil found across parts of East and Central Africa. When wet: vehicle sinks and slides unpredictably, traction near zero. When drying: surface sets to concrete hardness with deep ruts. Rule: if it is black cotton, assess whether the risk of crossing is acceptable. No technique substitutes for a dry track.

Sand-Mud Mix — MODERATE

Looser consistency, less suction than clay. Treads self-clean better. Often found on floodplain edges and river crossings. Key risk is hidden depth — the surface looks shallow but the vehicle can sink axle-deep unexpectedly. Probe with a stick before crossing any unfamiliar patch over 20 m long.

Tropical Laterite — VARIABLE

Red-iron soil common across many parts of Africa. Dry: excellent traction. Wet: surface becomes incredibly slippery — similar to oiled clay with a very thin grip window. Cross-camber on laterite roads in rain requires extreme speed reduction and deliberate line. Laterite dust also packs air filters rapidly when dry.

APPROACH AND LINE

- **Where to cross, not how fast.**
  Walk the crossing before committing the vehicle. Probe depth. Locate the highest, most compact ground. Entry and exit angles matter — a steep exit bank will sink the front axle into the far edge of the crossing.
- **Approach straight.**
  Angled entry to a mud crossing allows one side to sink faster than the other. Once a vehicle starts to lean in mud, recovery is very difficult. Straight in, straight out where the terrain allows.
- **Survey the exit before entry.**
  A passable entry with an impassable exit is a trap. Walk the whole crossing, not just the entry.
- **Tyre pressure.**
  In clay and black cotton, lower pressure is a double-edged sword — wider footprint but lower ground clearance and more suction. On most mud types, use R14's mud row. Do not go to sand pressures on mud.

THROTTLE AND STEERING DISCIPLINE

1. **Smooth, consistent throttle.**
   In mud, wheelspin is your enemy. The moment a tyre breaks traction in clay or black cotton, it excavates a hole that the vehicle sinks into. Apply power steadily and progressively — the turbo 1HZ's torque is an asset; its sudden onset is a liability in mud.
2. **Minimal steering corrections.**
   Every steering input pushes the front tyres sideways through the mud rather than forward. Once you have selected your line, hold it with the smallest corrections possible. Large corrections mid-crossing create resistance and slow you down critically.
3. **Do not stop.**
   Once you are committed and moving, finish the crossing. Stopping in mud to reconsider is rarely recoverable without assistance. If you must stop, deploy recovery gear immediately — do not allow the vehicle to settle deeper.

When you stop moving in mud — immediate response: Ease off throttle instantly. Do not spin wheels. Assess depth, suction, and available anchor points before your next move. Rocking (forward/reverse alternating) works briefly in light mud; in heavy clay or black cotton it deepens the hole rapidly. Deploy recovery boards or call for a snatch recovery. Cross-ref R9.

MUD AND THE 1HZ — INTAKE, COOLING, UNDERBODY

1. **Air filter risk.**
   The 1HZ's air intake path runs through the pre-cleaner and paper element. Fine mud spray — particularly laterite dust mixed with rain — bypasses the pre-cleaner and loads the main element rapidly. Check and service the air filter after any extended mud running. A restricted filter dramatically reduces power and increases EGT. Cross-ref R4, R18.
2. **Radiator blockage.**

Mud splatter packs the radiator core within a single crossing in thick conditions. Monitor coolant temperature continuously in mud work. At first sign of rising temperature, stop and clean the radiator face — flush with water, not pressure (the core fins bend easily). Cross-ref R22.

3 **Underbody packing.**
Clay mud sets hard under the vehicle. After a heavy mud section, inspect and clear: leaf spring pockets (impedes suspension articulation), diff breather tubes (check for mud packing), undercarriage generally. Do not let clay set before cleaning — it becomes concrete.

POST-MUD INSPECTION PRIORITIES

1 **Brakes.**
Mud packs between pads and rotors, causing temporary brake fade or noise. After a mud crossing, drag the brakes gently for 50 m at low speed to clear mud from the rotors. Check that brake feel returns to normal within 200 m.

2 **Diff breathers.**
Front axle, rear axle, and transfer case breathers can become submerged or packed. Check that they are clear and unobstructed. If any show signs of mud ingress, check diff oil immediately for water contamination (milky appearance).

3 **Front hubs.**
Even with manually-locked hubs, mud can work into the hub seal interface. Cross-ref R13 drivetrain section for hub inspection procedure after heavy mud work.

4 **Air filter.**
Check pre-cleaner and main element. Tap out, inspect, replace if in doubt. On expedition, carry at least one spare main element.

5 **Underbody pack.**
While vehicle is still accessible and mud is still wet, clear spring pockets, shock lower mounts, and any packed material from brake components.

CROSS-REFERENCE R9 Vehicle Recovery R13 Drivetrain R22 Cooling System R27 Bush Mechanics

3 — ROCK / TECHNICAL TERRAIN

Rock and technical terrain rewards patience and penalises urgency more than any other surface. The 76 SW's long wheelbase is its primary liability here — the break-over angle is reduced compared to a short-wheelbase vehicle, and the rear overhang can strike on exit ramps. Speed is measured in metres per minute, not km/h.

TYRE PRESSURE ON ROCK

Lower pressure increases sidewall flex and ground contact — but too low on sharp rock creates sidewall cuts that go through the carcass. The 265/75R16 AT has a reasonable sidewall height for this compromise.

ROCK TYPE	R14 ROW TO USE	RATIONALE
Hard smooth rock / ledge work	Rocky terrain (crawl)	Sidewall protection priority. Smooth surface means grip is not tyre-footprint limited. Maintain structural integrity.
Broken rock / boulder field	Rocky terrain (crawl)	Some flex helps the tyre contour irregular surfaces, but do not go below this row on sharp angular rock with the 76's weight.
Rocky track / corrugated stone	Gravel (firm), or Corrugated dirt where it is corrugated	Sidewall protection on square-edged stone predominates. Lower pressure gives more pinch-flat risk than traction benefit. Keep to that row's speed ceiling.

On sharp rock, never below R14's rock-crawl row. Lower than that, the 265/75R16 AT sidewall flexes enough to contact the rock face at the tyre bead area. A bead bruise does not fail immediately — it fails 20 km later under load. Keep the compressor ready so that when soft terrain gives way to rock you can bring the pressures straight back up to the right R14 row.

WHEEL PLACEMENT — TYRE, NOT VEHICLE

PRINCIPLE 01

ONE WHEEL AT A TIME

Plan each wheel placement individually before moving. Identify where each tyre will land before applying power. In a boulder field, the right line is the one where three wheels are stable while the fourth crosses the obstacle — not the one that looks straightest from the driver's seat.

PRINCIPLE 02

DRIVE THE TYRES

Your point of focus should be the tyre contact patch, not the bonnet. Where the tyre sits determines everything: clearance, stability, articulation. Develop the habit of memorising exactly where your tyres are relative to obstacles.

PRINCIPLE 03

USE THE SUSPENSION

The OME medium lift setup on this vehicle is designed for cross-axle articulation. Let it work. Keeping three tyres on the ground while the fourth lifts is normal and correct. Do not rush the crossing — give the suspension time to articulate before applying more power.

PRINCIPLE 04

MOMENTUM IS MINIMAL

Rock driving speed is near-walking pace. The goal is control, not speed. Momentum helps only in rare specific situations (climbing out of a deep step). For all other rock work, creep through under power — the brakes are for stopping safely, not for slowing on the obstacle.

RIDGES, DITCHES, RUTS AND AXLE-TWISTERS

- **Ridges square on.**
  Both wheels of an axle go over together; at an angle one wheel lifts and that axle loses drive (St Pierre White 2014, p. 190).

- **Ditches, troughs and gullies at an angle — never square on.**
  About 45°, so only one wheel at a time drops in and three stay on the ground. Square on, the nose digs in, the rear overhang catches or the vehicle hangs cross-axled; too sharp an angle risks a roll. Leave a V-gully at an angle too (Cobley 2013, ch. 12; St Pierre White 2014, pp. 190–191; Sheppard 2008, §5.3-21).

- **Deep ruts: straddle.**
  Where the middle would touch the diffs, run the wheels on the ridges or one wheel in and one out, slowly (owner ruling, 12 Oct 2026; St Pierre White 2014, p. 191). In shallower ruts, keep the front wheels straight and let the vehicle steer itself — turned wheels add drag and damage sidewalls (Cobley 2013, ch. 11). Probe water-filled ruts with a stick first.

- **Axle-twisters — push and pull.**
  Low range first or second, diff locks in. Ease the throttle as each front wheel drops back so it lands without spinning and grips, then feed power in again as it pulls. Too much speed bounces the wheels off the ground (St Pierre White 2014, p. 191).

- **Cross-axled and spinning?**
  Lock the diffs first. If still stuck, dig under the loaded wheel diagonally opposite the spinning one — that lowers the vehicle back onto the airborne wheel — rather than packing under the spinner (St Pierre White 2014, p. 215; Sheppard 2008, §5.3-21).

SIDE SLOPES

A side slope is where a vehicle is most likely to roll. If there is another way round, take it. Fear usually arrives before the real danger — obey it and back out early (Cobley 2013, ch. 13; Sheppard 2008, §5.3-26).

- **Walk it first.**
  Look for holes, roots, rocks and soft spots that would suddenly add lean; a downhill wheel dropping into a hole is how rolls start (Cobley 2013, ch. 13; Allen 2021, 'Essential Skills').

- **Passengers out, walking well clear.**
  Nobody on the downhill side, and nobody riding on or hanging off the vehicle (owner ruling, 12 Oct 2026; Cobley 2013, ch. 13). Sheppard has helpers hang on the uphill side to add weight (2008, §5.3-26); the library does not adopt it.

- **Secure the load.**
  Anything that slides downhill adds to the lean. Keep weight off the roof rack (Cobley 2013, ch. 13; St Pierre White 2014, p. 186).

- **Low range, first, diff locks in, slow and steady.**
  Steering as straight as possible. Sit square in the seat — leaning into the slope only confuses your sense of the real angle (Cobley 2013, ch. 13; Allen 2021, 'Essential Skills').

- **If it starts to slide or tip: steer downhill.**
  Add a touch of throttle if needed. Never steer uphill — if the uphill wheels suddenly grip, the vehicle rolls (Cobley 2013, ch. 13; Sheppard 2008, §5.3-26; Allen 2021, 'Essential Skills').

- **Windows closed, arms inside.**
  The instinct in a roll is to put an arm out; it gets crushed (Cobley 2013, ch. 13).

- **Never open the downhill door on a leaning vehicle.**
  Its weight can tip the balance. Get out on the uphill side (Cobley 2013, ch. 13).

DIFF LOCKS — WHEN TO ENGAGE AND DISENGAGE

1. **Engage before the obstacle, not on it.**
   The transfer case and front/rear diff locks must be engaged and confirmed before you enter technical terrain. Attempting to engage a diff lock when one wheel is spinning or airborne can damage the locking mechanism. Stop, engage, confirm light, then enter.

2 **Disengage before returning to open terrain.**
A locked diff on a flat, grippy surface creates understeer and places severe stress on the drivetrain when cornering. The 76's front diff lock in particular must be disengaged before returning to maintained tracks. Cross-ref R13.

3 **Part-time 4WD — no centre differential lock (CDL).**
This vehicle has no CDL. In 4WD on a grippy surface, transmission wind-up occurs. Never use 4H on tar, concrete or any hard, grippy surface — return to 2H when the surface firms up. 4H is right on loose, corrugated or rough gravel: it steadies the vehicle and spreads shock loads over both axles (owner ruling, 12 Oct 2026; Sheppard 2008, §5.3-10; St Pierre White 2014, p. 192). If the steering stiffens and the vehicle resists a turn, that is wind-up — return to 2H. Cross-ref R13.

APPROACH, BREAK-OVER, AND DEPARTURE ANGLES — LC76 SW

LC76 SW geometry: The long wheelbase (2,730 mm — R25) makes the break-over angle the tightest of the three; the rear overhang and the tow-bar receiver cut into the departure angle. The break-over angle is the critical constraint — the belly between the axles will ground on any obstacle the clearance geometry suggests should be passable. When in doubt, use a spotter for break-over assessment before crossing step obstacles.

SPOTTER SIGNALS

▸ **Spotter positions beside and slightly behind the driver's door** — good sight line to front obstacle and rear departure. Never directly behind the vehicle.

▸ **Come forward:**
Open hand, palm up, beckoning gesture. Speed control by gesture pace — slow beckoning = creep.

▸ **Stop immediately:**
Fist raised and held. Non-negotiable — driver responds immediately without question.

▸ **Steer left / steer right:**
Flat hand pointing in direction. Degree of urgency shown by speed of gesture and distance of point.

▸ **Reverse:**
Both hands, palms facing driver, pushing backward motion.

▸ **OK / proceed without guidance:**
Both thumbs up. Spotter steps clear.

High-centering: The 76 SW is at genuine risk of high-centering on sharp step obstacles due to its reduced break-over angle. Prevention: never assume clearance — use a spotter for any step you cannot see clearly past from the driver's seat. Response once high-centred: do not spin wheels, assess which axle is the primary support point, lift the grounded end with the hydraulic jack on its base plate (bottle jack as backup stand), pack under the wheels and lay the recovery boards — or dig the high point away. No hi-lift is carried. Cross-ref R9, R40 §2.

CROSS-REFERENCE R9 Vehicle Recovery · R13 Drivetrain Training · R14 Tyre Management · Tyre Pressure Calculator

4 — WATER CROSSINGS

A water crossing is one of the few off-road situations where a single mistake can result in total vehicle loss rather than a recoverable bog. The decision to cross requires discipline: the vehicle is not built for submersion, and ego has drowned more overlanders' equipment than any river. Walk it first. Always.

ASSESSMENT BEFORE ENTRY

Q1
What is the depth?

Limit: the Safari snorkel puts the engine air intake above the roofline, so the practical limit is set by the door seals, the standard (unextended) gearbox and transfer-case breathers and the electrics — see the depth box below. Walk it with a stick. At vehicle-width, check mid-channel depth, not just edges. Deep entry channel and shallow far side is still a crossing — assess the deep point.

Q2
What is the current?

Passable: You can stand in it without bracing hard.
Danger: Thigh-deep current that moves you = vehicle-moving force. Water moving a vehicle sideways at depth creates capsize or sweeping risk.

Q3
What is the bottom?

Rock: firm but tyre slip risk. Sand/gravel: usually firm unless fine silt. Clay/mud: can sink suddenly. Probe with stick for soft patches. Hidden obstacles — submerged rocks, tree limbs — are common. Walk the line.

PRE-CROSSING PREPARATION

Front hubs locked → Low range engaged → Diff locks set → Belts off, window down → Snorkel joints checked → Radiator blind on → Entry point confirmed

Belts off, driver's window down before deep water. If the vehicle stalls in deep water or the current takes it, the window is the way out — water pressure holds a door shut, and electric windows may fail once wet. Seatbelts stay on for steep and side-slope work; they come off for deep wading only (St Pierre White 2014, p. 172; Cobley 2013, ch. 14). This is not permission to enter a crossing where submersion is likely.

Radiator blind. A tarp or plastic sheet over the grille holds back the rush of water into the fan area (Sheppard 2008, §5.3-34). **Keep the belts on** — the fan is viscous-coupled and slips when it meets water, and the V-belt also drives the water pump and power steering (R4).

CROSSING TECHNIQUE

1. **Slow and steady — maintain bow wave.**
   Enter at a consistent pace that creates a small bow wave ahead of the vehicle. The bow wave reduces the effective water depth around the engine bay by 50–100 mm. A speed of 5–8 km/h is typically correct. Too slow = no wave protection; too fast = wave breaks over the bonnet.

2. **Do not stop mid-crossing.**
   Stopping kills the bow wave and allows water depth to equalise around the vehicle — including around the exhaust outlet and diff breathers. If you stop unavoidably, restart immediately and maintain momentum.

3. **Do not change gear mid-crossing.**
   Select second gear low range before entry. Maintain it throughout. A gear change mid-crossing interrupts power and can allow water ingestion through the exhaust on overrun.

4. **In a current — angle into it.**
   Never sit broadside to moving water. Choose the line from where the entry and exit lie, and as a rule angle into the current so the vehicle is not pushed downstream (owner ruling, 12 Oct 2026; Cobley 2013, ch. 14; Allen 2021, ch. 5). If you cannot walk against the flow without bracing hard, do not drive it (St Pierre White 2014, p. 190).

1HZ-SPECIFIC: AIR INTAKE AND DEPTH LIMITS

Snorkel fitted (Safari, Build Register). The engine breathes from above the roofline, so the intake is not the first thing water reaches. Water in the engine still means hydraulic lock and bent or broken rods — not recoverable in the field — so check the snorkel body, its A-pillar mount and every joint and clamp before deep water (R4).

Practical crossing depth: about 800 mm (water at door-sill level), depending on speed and bow wave. What limits it is not the intake but the door seals, the electrics and the gearbox and transfer-case breathers, which are standard (only the diff breathers are extended). Above about 400 mm, check the gearbox and transfer oils as well as the diffs afterwards. *The 800 mm figure is carried in R1 and R11 and is being traced to a published source (OI-116).*

POST-CROSSING INSPECTION

1. **Brake drag to dry rotors.**
Immediately after exit, apply light sustained brake pressure for 100–150 m to dry and clean the brake rotors and pads. Check that brake feel returns to full normal within 300 m.

2. **Diff and hub inspection for water ingress.**
After any crossing above 400 mm, check differential oil for water contamination (check plug — milky or foamy oil is contaminated). Front hubs: check for water in the hub cavity. Cross-ref R13.

3. **Air filter inspection.**
Any crossing where bow wave exceeded bonnet height — check the air filter immediately. Water in the filter element restricts airflow and can allow moisture into the intake tract.

4. **Service history trigger.**
Any deep crossing (above 400 mm) should be logged in R8 Service History as a maintenance trigger: diff oil check, brake inspection, hub inspection at or before next scheduled service.

Fail states — know these before you enter:
Stalled mid-crossing: If there is any chance water reached the engine — a stall as the bow wave broke over, a cough, a sudden stop — do NOT restart: turning over a hydraulically locked engine destroys it. Winch or snatch the vehicle out, then deal with the engine on dry ground. Only if the stall clearly had another cause and the intake stayed dry, try one restart (Cobley 2013, ch. 14; St Pierre White 2014, p. 188). Cross-ref R9.
Swept vehicle: If the current starts moving the vehicle sideways, get out through the open window on the downstream side — water pressure holds the upstream door shut. Do not fight the current from inside the vehicle.
Door seal failure: Water entering the cabin rapidly — exit via window if doors cannot be opened due to pressure. Practice window-exit drill before needing it.

CROSS-REFERENCE R8 Service History R9 Vehicle Recovery R13 Drivetrain Training

5 — CORRUGATIONS AND GRAVEL ROADS

Corrugations are the daily reality of remote overland travel — not a dramatic terrain challenge, but a relentless mechanical stress test that loosens fasteners, fatigues components, and degrades the vehicle over thousands of kilometres. They are also the terrain type most likely to be underestimated.

THE PHYSICS OF CORRUGATION

1. **Resonant frequency trap.**
Corrugations form at the resonant frequency of the average vehicle's suspension. At low speed, the vehicle bounces into every corrugation. There is usually a 'harmonic' speed — typically 40–70 km/h — where the body rides more smoothly. It is a comfort band, not a safe one: the axles and springs still take a beating, and because the tyres are skipping from crest to crest, grip for braking and steering is badly reduced. Go no faster than you need to reach it, look further ahead and be gentle with brakes and steering (owner ruling, 12 Oct 2026; Sheppard 2008, §5.3-33; Sheppard 1994, §4.9; Cobley 2013, ch. 10).

2. **Finding the float speed.**
The OME medium-lift setup on this vehicle (coils front, leaf rear) shifts the float speed slightly compared to standard. Experiment on the first corrugated section to find the speed where harshness reduces. Find it within the 40–70 km/h band and never above the R14 speed ceiling for the pressure you are running. Conditions vary: a loaded vehicle rides differently from an unloaded one.

3. **When float is not possible.**
On severely corrugated tracks or where speed is restricted by curves, terrain, or dust, slow down to below 30 km/h and pick a line — often the edge of the track has fewer corrugations than the centre. Accept the rough ride. Do not try to float in marginal conditions where speed itself creates hazard.

LOAD DISTRIBUTION EFFECT

The rear leaf spring setup reacts to corrugations differently at different load levels. An unloaded 76 with an empty roof and bare cargo bay will bounce more harshly than an expedition-loaded vehicle because the leaves are stiffer relative to the weight. The OME CS058R rear leaf is rated for medium-heavy loads — at full expedition load (~2,450 kg actual), the suspension is in its operating range and corrugation behaviour improves. Position heavier, denser cargo low and rearward for best corrugation compliance.

FASTENER AND COMPONENT FATIGUE — KNOWN LC76 VULNERABILITIES

Exhaust System — HIGH FATIGUE

The factory and aftermarket exhaust hangers crack at the weld points over extended corrugation exposure. The Powerflow system uses better hanger material, but check all hanger points after 500 km of corrugated driving. Exhaust noise increase is the primary symptom.

Body Mounts & Bolts — HIGH FATIGUE

The 76 body-to-chassis mounting bolts loosen progressively on corrugations. Include all body mount bolts in the post-corrugation inspection. Loose mounts cause body creak, misaligned doors, and eventual fatigue cracking around the mount points.

Shock Absorber Mounts — MODERATE

Upper and lower mount bolts (OME Nitrocharger Plus front and rear) work loose on sustained corrugations. Check torque on both front N181 and rear 63111 mounts after 500 km corrugated sections. A loose shock mount sounds like a rhythmic knock on rough surfaces.

Electrical Connections — VARIABLE

Vibration fatigues crimp connections, Deutsch connector retention springs, and Victron-to-battery terminal connections. After 1,000 km of significant corrugation, inspect all electrical connections in the cargo area — particularly the LiFePO4 battery terminals and BMS connections.

Wheel Nut Torque — CRITICAL

Wheel nuts back off on corrugations — more so at float speed where vibration frequency couples with the lug nut thread pitch. Check wheel nut torque every 500 km on extended corrugated sections — the wheel-nut torque in R40 §6 (also on the R1 card; being verified against the handbook — OI-7). A loose wheel sounds like a rhythmic rumble or whump at speed.

Roof Rack Mounting — MODERATE

Any roof-mounted rack on the 76 creates a high centre-of-gravity vibration amplifier. Roof rack attachment bolts — to the roof rail and gutter — fatigue progressively. Check all rack mounts at each post-corrugation inspection. A loose rack damages the roof rail and can eventually deform the roof skin.

TYRE PRESSURES ON CORRUGATIONS

Corrugation pressure: use the R14 'Corrugated dirt' row — it is the master. A moderate drop from road pressure softens the hammering on the vehicle, provided speed stays within the ceiling for that pressure (Sheppard 2008, §5.3-33; Cobley 2013, ch. 10). Pressure and speed go together: lower pressure, lower speed.

GRAVEL ROADS — DUST, TRUCKS, BERMS AND DIPS

- **Oncoming truck in a dust cloud.**
  Slow to a crawl and get well over, or pull off and stop. Stones fly, and another vehicle may be overtaking inside the cloud (St Pierre White 2014, p. 194).
- **Inside the dust, let the engine idle.**
  Where it is safe to coast, dip the clutch and let the revs fall to idle until you are through — the engine draws in roughly a quarter of the dust it otherwise would — then bring the revs back and let the clutch in gently (Sheppard 2008, §5.3-35). Hang back rather than drive in another vehicle's dust; in a dust storm, stop and switch off. Headlights on so others can see you (St Pierre White 2014, p. 195). Air filter: R4.
- **Watch the berm when overtaking.**
  Outside wheels in the loose gravel at the road edge drag and can spin the vehicle, and a steep camber tilts it as you move over (St Pierre White 2014, p. 194). Diff locks off on gravel roads (St Pierre White 2014, p. 194).
- **Dip or washaway seen too late.**
  Brake hard without locking the wheels, and release the brakes just before the front wheels hit — the nose lifts and the front axle takes far less. Never swerve into it sideways (St Pierre White 2014, pp. 194–195).

BLOW-OUT AT SPEED

No hard braking. Take your feet off the pedals, hold the wheel straight, keep away from the camber, and change down one gear gently as the vehicle slows; then stop well off the road. Hard braking in a blow-out leads to a slide, and a slide into even a shallow rut can roll the vehicle. A rear blow-out is still hard to control. 4H on gravel cuts the loss of control a great deal (St Pierre White 2014, p. 197). Wheel change: R40.

POST-CORRUGATION INSPECTION CHECKLIST

1. **Wheel nut torque — all five per wheel.**
   To the R40 §6 torque (R1 card; OI-7). Use torque wrench, not feel.
2. **Exhaust hanger visual — all mounting points.**
   Check for cracks, detachment, rubber deterioration.
3. **Shock mount bolts — front and rear, both sides.**
   Check torque. Creaking or knocking on bumps = first symptom.
4. **Body mount bolts — accessible points.**
   Front and rear body-to-chassis mounts particularly.
5. **Roof rack mounts — all attachment points.**
6. **Electrical connections — Victron battery terminals and BMS.**
7. **General cargo bay walkthrough — what has moved, shifted, or come loose.**

6 — ALTITUDE

The route crosses significant altitude changes — from sea level coastal sections to plateau elevations above 3,000 m. Altitude affects the 1HZ's performance, the turbocharger's operating envelope, cold-start behaviour, and the cooling system's ability to maintain safe temperatures.

1HZ POWER LOSS AT ALTITUDE

Naturally-aspirated baseline: The 1HZ loses approximately 3% of power per 300 m of altitude above sea level due to reduced air density. The SAC low-boost turbocharger partially compensates — but turbo boost pressure also falls as air density decreases. At 3,000 m altitude, expect power to be noticeably reduced — reserve your assessment of the engine's condition for sea-level operation. What feels like an engine problem at altitude may simply be altitude derating. Cross-ref R4, R18.

EGT BEHAVIOUR AT ALTITUDE

1. **EGT rises at altitude under equivalent load.**
   Less oxygen per unit volume means the injected fuel-to-air ratio increases for a given throttle position. The engine runs richer at altitude without any change in driver input, which raises exhaust gas temperature. If you have an EGT gauge, establish your sea-level normal range first and apply an altitude correction of approximately +20–30°C per 1,000 m under equivalent load. Cross-ref R21 Turbo Operations.

2. **Manage EGT proactively at altitude.**
   On steep climbs above 2,000 m, reduce load by selecting a lower gear earlier than you would at sea level. This keeps engine speed higher and fuelling lower for a given road speed, reducing EGT. Do not sustain maximum throttle on long climbs at high altitude — rest the engine on descents.

DIESEL QUALITY AT ALTITUDE

Diesel quality at remote high-altitude fuel depots can be variable — older or degraded stock is more common the further you are from major supply routes. Poor-quality diesel burns less efficiently, raises EGT further, and can produce visible black smoke under load at altitude even from a healthy engine. If the vehicle produces heavy black smoke on a grade and clears at lower altitude, consider fuel quality as the variable. Cross-ref R24 Fuel System Guide.

COLD STARTING AT ALTITUDE

1. **Glow plug discipline — full cycle.**
   At altitude above 2,000 m with overnight ambient temperatures below 5°C, the glow plug warm-up cycle is critical. Allow the full glow indicator to complete and hold for a further 3–5 seconds before cranking. The 1HZ requires a successful compression ignition — thin air and cold temperatures reduce the margin available for a marginal glow plug. Cross-ref R4, R18.

2. **Wait — do not crank repeatedly.**
   If the engine does not fire on the first crank, wait 30 seconds before the second attempt. Repeated cranking without waiting floods the cylinders with unburnt fuel, contaminating the oil and making the next start attempt harder. Two full glow cycles if necessary. Cross-ref R18 for non-start diagnostic sequence.

MOUNTAIN PASS TECHNIQUE

1. **Engine braking on long descents — gear selection.**
   Select the same gear you would use to climb the grade. On a long 10% descent, if you climbed in 2nd low range, descend in 2nd low range. The R151F and transfer case provide engine braking that spares the brakes on sustained grades. Avoid riding the brakes continuously — brake fade on a mountain pass is a serious emergency.

2. **Brake fade prevention.**
   Apply brakes firmly to scrub speed, then release fully to allow cooling. Do not drag brakes. If brake pedal begins to feel soft or spongy on a long descent, stop immediately, allow 5–10 minutes of cooling with wheels stationary, then continue with more aggressive engine braking. Cross-ref R10 Emergency Procedures.

COOLING SYSTEM AT ALTITUDE

Boiling point drops with altitude: At sea level, coolant with a 50/50 mix boils at approximately 129°C (with a 1.0 bar pressure cap). At 3,000 m altitude, atmospheric pressure is reduced, and the system boiling point drops by approximately 9–12°C. A borderline cooling system that manages at sea level can overheat on a long Alpine-style climb at altitude. The pressure cap's condition is critical — a weakened cap that vents early reduces the boiling point further. Check pressure cap condition before any high-altitude section. Cross-ref R22 Cooling System Guide.

CROSS-REFERENCE R4 Engine Training · R10 Emergency Procedures · R18 1HZ Troubleshooting · R21 Turbo Operations · R22 Cooling System · R24 Fuel System Guide

7 — NIGHT DRIVING

The default position for overland travel in remote areas is: do not drive after dark. This is not timidity — it is risk management. In most situations, the risks of driving at night outweigh any schedule benefit. This section defines when the rule applies, when exceptions are warranted, and how to manage night driving when unavoidable.

WHEN TO DRIVE AT NIGHT — AND WHEN NOT TO

HIGH RISK — DO NOT DRIVE

Rural roads after 21:00. Unfamiliar road. Near major waterhole or park boundary. Livestock country. No GPS waypoints loaded. Fatigued driver or co-driver. Any technical terrain ahead.

MODERATE RISK — PROCEED WITH CAUTION

Known road, well-lit town-to-town section. GPS waypoints confirmed. Both occupants alert. Speed under 80 km/h. Clear weather, good visibility. Defined destination within 60 minutes. Stopping points identified on route.

LOWER RISK — ACCEPTABLE IF REQUIRED

Paved well-trafficked road. Urban or peri-urban context. Daylight departure delayed unavoidably. Short distance to known safe stopping point. Both occupants involved and alert. Agreed stopping criteria in place.

The rule: Do not drive after dark unless the risk of stopping outweighs the risk of driving. An empty fuel tank, a dangerous stopping environment, or an impending border closure may all justify night driving. But "we want to make the next campsite" does not. If in doubt, stop before dark and resume at dawn.

LIGHTING

1. **Factory headlight output.**
   The 76 series factory H4 headlights provide adequate but not generous illumination at speeds over 70 km/h. At 100 km/h on an unfamiliar road in darkness, stopping distance may exceed visible range. If night driving is a regular expectation, supplementary lighting is appropriate.

2. **Supplementary lighting philosophy.**
   Driving lights mounted at bumper height give better road surface illumination than roof-mounted lights, which produce a flat, shadow-reducing light. Roof lights are useful for slow navigation and camp work, not high-speed night driving. Do not over-light to the point where you feel confident exceeding safe speeds.

ANIMAL STRIKE RISK

Livestock (cattle, donkeys, goats) and large game near waterholes are the primary night driving hazard in remote areas. Animals on dark roads are often not visible until they are within 30–50 m. A speed of 60–70 km/h at night in rural livestock country gives adequate stopping distance; 100 km/h does not. Hitting a large animal at speed is a severe accident — vehicle-totalling, injury-causing. Slow down after dark in animal country. There is no exception.

NAVIGATION AT NIGHT

- **GPS-only navigation after dark.**
  Map-reading at night requires interior light, which destroys night vision. Load waypoints and route before sunset. Use voice guidance or a co-driver managing the GPS display. The driver's eyes remain on the road. Cross-ref R16.

- **Pre-set the return-to-road waypoint before entering any off-road night section.**
  If night driving is forced by circumstance mid-track, mark your current GPS position before the track begins so you can navigate back if necessary.

- **Do not use internal map light while driving.**
  Even brief interior lighting destroys night adaptation. Allow 10 minutes for full night vision recovery.

FATIGUE MANAGEMENT

1. **The 22:00 rule.**
   After 22:00, the default position is stop unless there is a specific and compelling reason to continue. Fatigue impairment at 22:00 after a full day of driving is equivalent to low-level intoxication. This is not negotiable on the expedition. Agree it before departure.

2. **Co-driver role at night.**
   The co-driver is the driver's dedicated safety system after dark: navigating, watching for animals and obstacles, monitoring fuel and time, and monitoring the driver for fatigue signs. The co-driver does not sleep until the vehicle is stopped for the night.

3. **Stopping criteria.**
   Pre-agree: the co-driver has authority to call a stop at any time without discussion. The driver does not argue. Agreed stopping criteria might include: any sign of driver microsleep, inability to follow GPS directions, two missed turns, eye fatigue assessment by co-driver. Once called, it is called.

CAMP SELECTION WHEN FORCED TO STOP IN DARKNESS

- **Visibility.**
  Choose a position visible from the road — you may need assistance, and a completely hidden vehicle is harder to find. But do not camp on the road itself or immediately adjacent to the driving line.
- **Elevation.**
  If possible, slightly higher than surrounding ground — reduces cold air pooling, improves tent drainage, and keeps you above flash flood run-off lines.
- **Drainage.**
  Do not camp in a dry riverbed, watercourse, or obvious drainage channel regardless of how dry conditions appear. Flash floods travel at night with no warning.
- **Proximity to road.**
  Close enough for security visibility, not so close that passing vehicles (headlights, noise) disturb rest or that a road incident involves your camp. 20–50 m is typically appropriate.

CROSS-REFERENCE R16 Comms & Navigation R10 Emergency Procedures

8 — STEEP UPHILL

Steep climbs demand commitment, correct gear selection before the slope, and a clear plan for what to do if momentum is lost. The 1HZ's turbo lag and the R151F's gear ratios mean that arriving at a climb in the wrong gear is not recoverable mid-face. Preparation is everything.

BEFORE YOU CLIMB

1. **Select gear before the base of the climb.**
   Identify the correct gear while you still have momentum on flat ground. On a steep loose slope, a gear change mid-climb stalls the vehicle. As a starting point: 2nd or 3rd low range for loose/rocky steep terrain; 1st low range for very steep or severely rutted faces. The turbo 1HZ has power in the mid-rev range — do not lug it in too high a gear.
2. **Walk the line if uncertain.**
   A steep climb that is not clearly passable from the base should be walked first. Check: surface traction (loose rock, soft soil, wet grass — each behaves differently), rut depth, crest shape (blind crests that drop sharply on the far side), and exit at the top. A passable climb with an impassable exit is a trap.
3. **Diff locks: engage before the slope.**
   If the surface is loose, wet, or rocky, engage rear diff lock (and front if required) before the base of the climb. A spinning wheel mid-climb on a steep face causes immediate loss of momentum and directional control. Stop on flat ground, engage, confirm, then climb.
4. **Tyre pressure appropriate for surface.**
   Sand uphill: deflated to sand target. Rock uphill: R14's rock-crawl row. Mud uphill: standard deflated pressure. Attempting a steep mud or sand climb on road pressures dramatically reduces available traction.

CLIMBING TECHNIQUE

1. **Commit fully — smooth, sustained throttle.**
   Hesitation on a steep climb wastes the momentum that carries the vehicle through the hard section. Apply throttle progressively before the slope steepens and hold it. The turbo 1HZ responds well to consistent demand — do not feather the throttle on the face. Keep revs in the working range (1,600–2,400 rpm) where torque is available.
2. **Straight up the fall line where possible.**
   Any angle off the fall line on a steep slope creates a lateral lean. The 76 SW's high centre of gravity (roof tent loaded, tall suspension) makes a traversing angle on steep loose ground a rollover risk. Where the terrain forces an angled approach, assess the lean angle critically before committing.
3. **Crest with caution — do not crest blind at speed.**
   Reduce speed as the bonnet reaches the crest. The far side may drop sharply, have a soft landing, or be occupied. Approach the last few metres of any blind crest slowly enough to stop if required.

IF YOU LOSE MOMENTUM MID-CLIMB

Losing momentum mid-climb is an emergency situation. The vehicle is on a steep slope with no forward motion. Your options narrow immediately — act in the correct sequence:

1. Apply the footbrake firmly and hold the vehicle stationary. Do not let it roll backward uncontrolled.
2. **Do not touch the clutch.** Pressing it disconnects the engine braking that is holding you, and the vehicle can roll back faster than you can brake. If the engine stalls, let it — a stalled engine in gear holds the vehicle (St Pierre White 2014, p. 178–180; Basham, 'Stall starts'; Sheppard 2008, §5.3-24).
3. Back down, do not restart on the face. Hold on the footbrake, select low-range reverse and let the clutch out; ease off the brake and let engine braking take the vehicle straight back down, both hands on the wheel, steering on the mirrors. Retry from the bottom: if you lost traction, try a gear higher or a little more speed; if you ran out of power, a gear lower (St Pierre White 2014, p. 180; Cobley 2013, ch. 8).

4 If a second attempt is not viable — controlled reverse descent is the exit. See Section 9 (Steep Downhill) for reverse descent procedure. Do not attempt to turn around on a steep face.

! **Do not attempt a U-turn on a steep or loose slope.** A turning vehicle on a steep face is at extreme rollover risk. Reverse down in a straight line is always safer than turning.

IF THE ENGINE STALLS MID-CLIMB

! **Hold it on the footbrake — hard.** With the engine off there is no brake servo, so the pedal needs far more effort. The footbrake holds all four wheels; the handbrake holds only the rears, so use it only as an extra hold while you change gear (Cobley 2013, ch. 8).

1 Footbrake on. Clutch in only long enough to select **low-range reverse**, then let the clutch fully out. Ease off the brake until the vehicle sits held by engine compression alone. Front wheels straight; look behind.

2 Hands firmly on the wheel — in reverse the castor works the wrong way and can snatch the wheel to full lock. Turn the key: the 76 cranks in gear (owner-confirmed), the engine fires and the vehicle backs down under engine braking. A brief turn of the key is enough. If the engine is cold, glow first (Sheppard 2008, §5.3-24; Basham, 'Stall starts').

3 Feet off the clutch and accelerator. Steer straight back down the way you came, on the mirrors. If it runs away, brake to a stall and start again. Switch the key only to ACC, never LOCK, if you stall again — the steering lock can engage (Basham, 'Stopping, starting and stalling').

4 At the bottom, work out why: too high a gear (too little power) — go a gear lower; wheelspin (too little grip) — a gear higher or more run-up. Then retry from the bottom, never from mid-slope.

CROSS-REFERENCE R9 Vehicle Recovery R13 Drivetrain Training R18 1HZ Troubleshooting

9 — STEEP DOWNHILL

Steep descents are where vehicles are lost. Brake fade, loss of directional control, and rollover all occur on descents — rarely on climbs. The 1HZ and R151F provide strong engine braking that is the primary control tool. The brakes are for supplemental speed management, not primary descent control.

BEFORE YOU DESCEND

1 **Walk the descent if you cannot see the bottom.**
A blind descent is the most dangerous situation in technical driving. Walk to the bottom. Check surface condition, rut depth, any step drops mid-face, and the run-out at the base. Confirm you can stop safely at the bottom — a water crossing or soft bowl at the base of a steep descent is a trap.

2 **Select gear before the edge.**
Same gear you would use to climb the equivalent grade. On a 10–15% sustained slope: 2nd low range. On very steep faces (25%+): 1st low range. Select the gear on flat ground at the top, before the nose tips over. Do not change gear mid-descent on a steep or loose slope.

3 **Diff locks: rear engaged on loose or rutted descents.**
A locked rear diff prevents one rear wheel from spinning faster than the other if traction is uneven — which stabilises the rear of the vehicle on descent. Front diff lock on very steep technical descents where front wheel placement is critical. Disengage before returning to open terrain. Cross-ref R13.

DESCENT TECHNIQUE

1 **Engine braking is primary — brakes are secondary.**
In the correct low range gear, the 1HZ provides strong compression braking. This is consistent, fade-free, and uses no consumable. Keep the clutch fully engaged throughout the descent — do not coast in neutral or with the clutch depressed. Coasting removes all engine braking and sends the vehicle toward brake-only control.

2 **Straight down the fall line where possible.**
Same principle as uphill — any lateral angle increases rollover risk. On loose descents, a vehicle that begins to rotate sideways on the slope is very difficult to recover. If the terrain forces an angled descent, slow to a near-crawl before introducing any angle.

3 **Brake application: firm and release, not drag.**
If supplemental braking is needed beyond engine braking, apply the brakes firmly to scrub speed, then release fully to allow rotor cooling and tyre rotation to resume. Continuous light brake application (dragging) heats rotors rapidly and causes fade. On a long descent, intermittent firm application is safer than sustained light application.

4 **Do not brake in corners on loose surfaces.**
Braking while steering on a loose steep descent transfers weight forward and reduces rear traction exactly when rear stability is most needed. Scrub speed in the straight before the corner, not during it.

5 **If the rear starts to come round — a touch of throttle.**
On a loose descent the wheels can lock under engine braking and the rear try to overtake the front. Steer gently down the slope and feed in a little throttle to match wheel speed to road speed; it pulls the vehicle straight. Do not hit the brakes and do not press the clutch. On loose shale and dune faces, second low rather than first keeps the wheels rolling (St Pierre White 2014, p. 178–179; Cobley 2013, ch. 7; Sheppard 2008, §5.3-26).

BRAKE FADE — RECOGNITION AND RESPONSE

Brake fade on a descent is an emergency. Symptoms: pedal feels normal but vehicle does not slow; pedal travel increases; burning smell. If this occurs on a descent:

! **Get onto engine braking first.** Faded brakes are overheated pads, shoes and fluid; braking harder makes it worse. Use firm, steady pedal pressure for what braking is left, and let the engine do the work (R34).

1. Downshift immediately if not already in lowest available gear. Maximum engine braking is the priority.
2. If the vehicle is still accelerating uncontrollably, steer toward any uphill escape — a bank, embankment, or side slope — to arrest speed through ground resistance.
3. Once stopped or slowed to safe speed, stop the vehicle and allow brakes to cool with wheels stationary for a minimum of 10 minutes before proceeding. Cross-ref R10.

REVERSE DESCENT — WHEN FORWARD IS NOT POSSIBLE

1. **When to reverse down.**
   If a climb attempt has failed and the top cannot be reached, reverse descent is the correct exit. It is also correct when arriving at the top of a descent that turns out to be impassable. Never attempt to turn a loaded 76 on a steep or loose face — reverse is always safer.
2. **Low-range reverse gives strong engine braking.**
   Reverse (4.220) is geared almost as low as first (4.313) on the R151F (R13), so low-range reverse holds the vehicle on a steep face much as first does going forward. Keep the clutch out and both hands firmly on the wheel — in reverse the steering can snatch to full lock (Sheppard 2008, §5.3-24).
3. **Use a spotter for reverse descents.**
   The driver has no clear visibility of the descent line while reversing. Mary-Ann spots from alongside — never directly behind the vehicle. Standard spotter signals apply (see Section 3). The spotter controls the line and calls stops; the driver controls speed.
4. **Maintain straight alignment.**
   Any steering input while reversing down a steep slope tends to amplify — a small correction becomes a large swing. Keep inputs minimal. If the rear starts to slide sideways, steer into the slide (toward the direction the rear is moving) with a small, immediate correction.

IF THE ENGINE STALLS MID-DESCENT

! **Apply footbrake and handbrake immediately.** A stalled engine on a descent removes all engine braking. The vehicle will accelerate if not held. Secure the vehicle before doing anything else.

1. Hold on the footbrake (no servo with the engine off — press hard). Stay in gear with the **clutch OUT**: compression then holds the vehicle. Do not press the clutch.
2. Restart in gear — the 76 cranks in gear (owner-confirmed). Glow first if cold. A brief turn of the key; the engine fires and engine braking resumes at once (St Pierre White 2014, p. 179).
3. Ease off the footbrake and continue the descent in the same gear, feet off the clutch, both hands on the wheel.
4. Investigate why the stall occurred before attempting any further steep terrain. Common causes: clutch slip under load, incorrect gear selection, throttle lifted suddenly mid-face. Cross-ref R18.

CROSS-REFERENCE R9 Vehicle Recovery · R10 Emergency Procedures · R13 Drivetrain Training · R18 1HZ Troubleshooting

10 — PRE-TERRAIN CHECKLIST

A 15-point audit to complete before entering any serious terrain — sand, mud, rock, water crossing, or sustained technical section. This is not a tick-box formality. It is the difference between a controlled entry and an uncontrolled one. The list takes under five minutes if the vehicle is properly maintained. Do it every time.

Rule: This checklist is completed while stationary, before the terrain begins — not at the first problem after entering. If the check reveals an issue, resolve it before entry.

1. **Tyre pressures confirmed and set for terrain type.**
   Check actual pressure with gauge — not by feel. Use Tyre Pressure Calculator for terrain-specific targets. Deflate before entry, not after the first soft patch. Cross-ref R14, Tyre Pressure Calculator.
2. **Front hubs: locked or free as appropriate.**
   Manual hubs: confirm engagement by feel and check (not just switch position). Locked for all 4WD terrain. Free for return to tar.

3 **Transfer case: 4L or 4H selected and engaged.**
Engage before terrain — not mid-section. Confirm indicator light where fitted. 4L for all technical terrain, deep sand, mud, rock. 4H for fast corrugated gravel where speed is maintained. Cross-ref R13.

4 **Diff locks tested: front and rear actuated and confirmed.**
Turn the wheel slightly against the lock engagement to confirm each diff lock is seated. A diff lock that activates the indicator but has not fully engaged will fail at the worst moment. Cross-ref R13.

5 **Recovery gear: accessible without unpacking.**
Recovery boards, snatch strap, shackles, and the ARB hydraulic jack within 30-second access. Not at the bottom of the cargo bay under all provisions.

6 **Front and rear tow points: clear and accessible.**
Snatch ring or shackle fitted. Not obstructed by a spare wheel, snorkel fitting, or cargo overhang.

7 **Fuel level: adequate to complete the section and return if necessary.**
Terrain driving consumes significantly more fuel than road driving — sand and mud can double consumption. If the section is remote and return is the only exit, you need sufficient fuel for two transits. Cross-ref Fuel Range Calculator.

8 **GPS waypoint set for re-entry point or camp.**
If the terrain section has a single exit, mark the entry point so you can navigate back. In featureless dune or bush terrain, disorientation is a real risk. Cross-ref R16.

9 **Communication check: satellite device on, known contact briefed.**
If entering a remote section, activate your Garmin inReach or equivalent. Confirm your last known contact knows your plan and expected exit time. Cross-ref R16.

10 **Time of day: can this section be completed before dark?**
If the terrain section takes longer than expected, will you be driving at night or camping in the terrain? Factor this before entry, not during. Cross-ref Section 7 (Night Driving).

11 **Weather observation: any indication of rain or change?**
Rain during a mud or water crossing section can make a passable route impassable in hours. Sand that is firm in dry conditions becomes heavily compacted or washed out in rain. Check sky and recent weather track before committing.

12 **Partner awareness: Mary-Ann briefed on section type and exit plan.**
Both occupants agree on what the section involves, what the risks are, and what the plan is if things go wrong — before entry. No surprises mid-section.

13 **Air filter condition: pre-cleaner checked.**
Especially before dusty, sandy, or laterite sections. A heavily loaded pre-cleaner before a long section is a risk. Clean it. Cross-ref R4.

14 **Coolant temperature normal before terrain entry.**
Entering slow technical terrain with a cooling system already at upper operating range leaves no margin for the increased load of low-speed high-throttle work. Allow the engine to normalise before low-speed technical work. Cross-ref R22.

15 **Compressor and inflation kit accessible.**
After terrain, tyres must be re-inflated before returning to road speeds. Compressor in reach, not packed under camp load. ARB dual, VIAIR, or equivalent — confirm it works before a long remote section. Cross-ref R14.

CROSS-REFERENCE R2 Departure Checklist R9 Vehicle Recovery R13 Drivetrain Training R14 Tyre Management R16 Comms & Navigation R22 Cooling System R27 Bush Mechanics Tyre Pressure Calculator Fuel Range Calculator